

Module-III Question Bank- Server-Side Scripting (PHP)

Part-A (2 mark)

1. Write a Laravel program to create a named route called home that returns the message "Welcome to Amir Raja's PHP World!".
2. What is Laravel? Mention any **two advantages** of Laravel.
3. Write the Laravel route to display Hello Laravel when the /welcome URL is accessed.
4. Differentiate between **Laravel and traditional PHP development**.
5. Write a Laravel route that returns the student view.
6. What is **Composer**? Why is it used in Laravel?
7. Write Blade code using @if to display Welcome Student when \$isStudent is true.
8. Write the command used to **create a new Laravel project** using Composer.
9. What is **Artisan** in Laravel? Mention any two Artisan commands.
10. What is a **Controller** in Laravel? Write the Artisan command used to create a controller.
11. Write a Laravel route that passes the value "Amir Raja" to the student view.
12. Write a Laravel POST route for the URL /student.
13. What is **template inheritance** in Blade? What is the purpose of @extends?
14. What is the purpose of the **routes directory** in a Laravel project?
15. Write the Blade syntax to display a variable named \$name.

PART-B (5 Marks Each)

1. Build a Laravel controller + Blade view with a form: the user submits a bracket string and the view reports whether it is valid.
2. Build a controller method protected by a custom Middleware that blocks the request if the submitted array contains duplicate values.
3. Explain the Laravel Directory Structure. Describe the purpose of important folders such as app, routes, resources, public, and config.
4. Write a Blade template to display the following:
 - A heading "Student Details"
 - A variable \$studentName
 - A simple if condition that displays "Active Student" if \$status is true.
5. What is Blade Template Engine in Laravel? Explain its advantages and basic syntax used in Blade templates.
6. Build a controller method that merges overlapping intervals from submitted data and passes the result to a nested Blade partial (table view).
7. Write a Laravel program to create a controller named UserController with a method index() that returns the message "User Controller Page" and call it using a route.
8. Build a controller method protected by a custom Middleware that blocks the request if the submitted array contains duplicate values.
9. Develop a PHP program to export book details from a MySQL database to an Excel file using PhpSpreadsheet. Create **index.php**, **export.php**, and a **library** database with a **books** table containing **id**, **title**, **author**, and **price**.
10. Build a Laravel route + controller method that accepts an array and a target via query parameters and returns the Two Sum indices as JSON.
11. Create a route with a parameter that runs binary search over a fixed sorted array on the server and returns the correct insert position.

12. Build a controller that moves all zeroes in a submitted array to the end and displays the before/after arrays in a Blade view using @foreach.

PART-C (10 Mark Each)

1. Explain the concept of Route Parameters in Laravel. How are parameters passed in routes? Give an example.
2. Write a Laravel program to:
 - a. Create a GET route /welcome.
 - b. The route should call a view named welcome.
 - c. Pass a variable \$name = "Amir Raja" from the route to the view.
 - d. Display the name in the view using Blade template expression.
3. Create a named route + controller that accepts a list of strings and returns their longest common prefix, rendered through a Blade view.
4. Write a Laravel program to:
 - a. Create a controller named StudentController.
 - b. Define a method show(\$id) in the controller.
 - c. Create a route /student/{id} that calls this method.
 - d. Display the message "Student ID is: {id}".
5. Explain the MVC (Model–View–Controller) architecture in Laravel with a neat diagram and example. Describe the role of Model, View, and Controller in application development.
6. Explain the steps to set up Laravel development environment. Discuss the role of Composer and explain how to create a new Laravel project.
7. **Develop a PHP-based Student Management System to export student data from a MySQL database into an Excel file using Composer.**

Requirements:

Create the following files:

- **index.php** – Create a web page containing a button “**Export Data to Excel**”.
- **export.php** – Connect to the MySQL database, retrieve all records from the students table, create an Excel spreadsheet using **PhpSpreadsheet**, and download the file as **Student_Details.xlsx**.
- **Database** – Create a database named **college** and a table named **students** with the fields id, name, email, and course. Insert at least three student records.

The exported Excel file must contain the columns **ID, Name, Email, and Course**, followed by all student records retrieved from the database.

Note: Use the **PhpSpreadsheet** library for generating the Excel file and configure the required HTTP headers so that the Excel file is downloaded when the “**Export Data to Excel**” button is clicked.

8. Build a Laravel Resource Controller endpoint that groups a submitted list of words into anagram groups and displays them in a Blade view.
9. **Programming Question**

Develop a PHP program to export employee details from a MySQL database into an Excel file using PhpSpreadsheet.

Create the following:

index.php – Display a button “**Export Employee Data**”.

export.php – Connect to the database, fetch employee records, and generate an Excel file named **Employee_Details.xlsx**.

Database – Create a database named **company** with an employees table containing **id, name, email, and department**. Insert at least three records.

The Excel file should display the employee details with proper column headings and should be downloaded automatically when the button is clicked.